

HIREON: ZERO-SURPRISE COST RISK AUDIT

Full Codebase Audit | June 2026 | Exchange Rate: \$1 = Rs 84

Scope: 1 Client | 4,900 Bulk Import + 100 AI Resume Parse/Month | Voice Answers Active (No STT)

SECTION 1: CRITICAL RISK FINDINGS FROM CODEBASE AUDIT

Discovered by reading actual source code. MUST be resolved before going live to avoid hidden expenses.

RISK 1 [CRITICAL] | CLOUDINARY STORAGE IS ACTIVE | File: app/services/storage_service.py Lines 13-86

FINDING: Your code uploads ALL resumes, JDs, logos, and avatars to Cloudinary (checked at startup). Free Tier = 25GB only. When limit hits, Cloudinary Starter = \$89/month (Rs 7,476). This single item destroys all profit.

FIX: Switch to Supabase Storage (already included in Pro Plan with 100GB free). Remove Cloudinary env keys from production .env.

RISK 2 [CRITICAL] | CELERY/REDIS HAS NO HOSTING | File: app/celery_app.py Line 9-10

FINDING: Celery requires Redis as broker for background tasks (interview feedback check every 5 mins). Render Starter does NOT include Redis. Without it, all background tasks fail silently in production.

FIX: Use Upstash Redis Free Tier (10k commands/day). Free for 1 client. Takes 5 minutes to configure. Just set REDIS_URL in Render env vars.

RISK 3 [HIGH] | ELASTICSEARCH WILL BREAK ON RENDER | File: app/services/elasticsearch_service.py

FINDING: ES connects to localhost:9200 which works in Docker locally but not on Render Starter. Full-text candidate search will fail. Good news: code already has a PostgreSQL fallback in search_service.py.

FIX: Ensure ES URL is set to an invalid/empty value in production so code falls back to PostgreSQL ilike search. Cost: Rs 0.

RISK 4 [MEDIUM] | OPENAI + MISTRAL API KEYS PRESENT | File: app/config.py Lines 42-44

FINDING: openai_api_key and mistral_api_key fields exist in config. If any code path triggers them (e.g. a model fallback or voice processing), unexpected charges will appear on your card without warning.

FIX: Set openai_api_key="" and mistral_api_key="" in production .env. Since client does not need STT or transcription, these keys must be empty.

SECTION 2: COMPLETE FEATURE TO SERVICE COST MAPPING

Feature	Service / API	Monthly Cost	Required	Risk Level
Backend Server (24/7 uptime)	Render Starter Plan	Rs 588	MANDATORY	None
Database + Authentication	Supabase Pro Plan	Rs 2,100	MANDATORY	None
File Storage (needs fix)	Supabase Storage (switch from Cloudinary)	Included	MANDATORY	HIGH if not switched
Voice Audio Storage	Cloudflare R2 (Free Tier)	Rs 0 - Rs 17	MANDATORY	None
Resume Parsing (100/month)	Groq API - Llama-3.3-70b	~Rs 420	MANDATORY	Low
JD Parsing	Groq API - Llama-3.3-70b	~Rs 50	MANDATORY	Low
AI Copilot Chatbot	Groq API - Llama-3.1-8b-instant	~Rs 200	MANDATORY	Low
Match Scoring + Summary	Groq API - Llama-3.3-70b	~Rs 200	MANDATORY	Low
HF Image Generation	HuggingFace Free (FLUX.1-schnell)	Rs 0	OPTIONAL	None
Email Notifications	Gmail SMTP (App Password)	Rs 0	MANDATORY	None
Interview Scheduling	Google Calendar API	Rs 0	MANDATORY	None
LinkedIn OAuth Login	LinkedIn Basic OAuth	Rs 0	MANDATORY	None
Push Notifications	Firebase FCM (Free)	Rs 0	OPTIONAL	None
Background Tasks	Upstash Redis Free Tier	Rs 0	MANDATORY	HIGH if not set up
Candidate Search	PostgreSQL Full Text (Supabase)	Included	MANDATORY	None
4,900 Bulk Import	Python Pandas (No External API)	Rs 0	MANDATORY	None
Analytics and Reports	PostgreSQL Queries (Supabase)	Included	MANDATORY	None
Frontend Hosting	Vercel Free Tier (Global CDN)	Rs 0	MANDATORY	None
Domain hirreon.com	Hostinger (Already Purchased)	Rs 0	DONE	None
GST 18% on Foreign SaaS	Indian Tax (Auto-added by Bank)	~Rs 484	MANDATORY	Often Missed

SECTION 3: MONTHLY COST SCENARIOS

A. MINIMUM MONTHLY COST (After All Critical Fixes Are Applied)

Service Item	Monthly Cost (Rs)
Render Starter Plan	Rs 588
Supabase Pro (DB + Auth + Storage)	Rs 2,100
Groq API (Resume Parse + Copilot + Scoring + JD)	~Rs 920
Cloudflare R2 for Voice Audio	Rs 0 - Rs 17
All Others: Gmail, Vercel, Firebase, Redis, Google API	Rs 0
GST 18% on Render + Supabase	~Rs 484
MINIMUM TOTAL	Rs 4,092 / month

B. RECOMMENDED PRODUCTION COST (With 20% Buffer for Spikes)

Service Item	Monthly Cost (Rs)
All infrastructure same as minimum	Rs 4,092
AI API buffer for unexpected usage spikes	Rs 500
RECOMMENDED TOTAL	Rs 4,600 / month

C. WORST CASE COST (If Critical Risks Are Left Unresolved)

Service Item	Monthly Cost (Rs)
Render Starter	Rs 588
Supabase Pro	Rs 2,100
Cloudinary Starter (if NOT switched to Supabase)	Rs 7,476
Bonsai.io Elasticsearch Paid Tier	Rs 1,260
Groq API with heavy usage spikes	Rs 1,500
Email Service Resend Pro (if Gmail blocked)	Rs 1,680
GST on all paid services	~Rs 800
WORST CASE TOTAL	Rs 15,404 / month

SECTION 4: ANNUAL PROJECTIONS AND CONFIDENCE SCORES

Annual Cost Summary

Scenario	Monthly (Rs)	Annual (Rs)
Minimum (All Fixes Done)	Rs 4,092	Rs 49,104
Recommended Production	Rs 4,600	Rs 55,200
Worst Case (No Fixes)	Rs 15,404	Rs 1,84,848

Confidence Score Per Cost Item

Cost Item	Estimate (Rs)	Confidence	Reason
Render Starter	Rs 588	HIGH	Fixed monthly plan. Zero variability. Confirmed on Render pricing page.
Supabase Pro	Rs 2,100	HIGH	Fixed monthly plan. Confirmed on Supabase pricing page.
Groq API Total	~Rs 920	MEDIUM	Depends on frequency of copilot usage by the recruiter. Can vary 20-30%.
Cloudflare R2	Rs 0 - Rs 17	HIGH	Mathematically confirmed. 250MB/month audio is within free 10GB for 3+ years.
GST 18%	~Rs 484	HIGH	Fixed statutory tax on all foreign SaaS bills in India.
Elasticsearch	Rs 0 (risky)	LOW	Depends on which code path activates in production. Needs manual verification.
Email Gmail SMTP	Rs 0	HIGH	Gmail SMTP is free. Google does not charge for SMTP with App Password.

SECTION 5: FUTURE COST TRIGGERS TO WATCH

What Can Increase Costs in Future and Under What Conditions

Trigger Event	When It Happens	Additional Monthly Cost
Cloudinary free limit hit (25GB)	Storage exceeds 25GB (approx 6-12 months)	+Rs 7,476/month
Groq rate limit exceeded	More than 100 parses per day consistently	Add API credits as needed
Supabase DB exceeds 8GB	After 2+ years of heavy platform use	Upgrade to Rs 4,200/month plan
Elasticsearch needed beyond PG	PostgreSQL search becomes too slow	+Rs 1,260/month (Bonsai.io)
Second client onboarded	When you add client number 2	+Rs 500/month (AI API only)
Email volume exceeds 300/day	Heavy notification blasts to candidates	+Rs 1,680/month (Resend Pro)

OpenAI key accidentally populated	Any audio/voice code path triggers it	Up to Rs 25,000/month
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SECTION 6: SALES SAFE PRICING RECOMMENDATION

Recommended Pricing Strategy for Client

Item	Amount (Rs)
Actual Operating Cost (Recommended Scenario)	Rs 4,600 / month
Minimum Safe Client Price (Break-even + buffer)	Rs 8,000 / month
Recommended Client Price	Rs 12,000 / month
Net Monthly Profit at Rs 12,000	Rs 7,400 / month
Annual Net Profit at Rs 12,000 (1 client)	Rs 88,800 / year
Profit Margin at Recommended Price	61%

SECTION 7: MANDATORY ACTION ITEMS BEFORE GOING LIVE

- [CRITICAL] Switch all file uploads (resumes, JDs, logos) from Cloudinary to Supabase Storage. Remove Cloudinary keys from .env.
- [CRITICAL] Configure Upstash Redis Free Tier and set REDIS_URL env variable on Render for Celery background tasks.
- [IMPORTANT] Set openai_api_key="" and mistral_api_key="" in production .env to prevent accidental API charges.
- [IMPORTANT] Verify PostgreSQL search fallback activates correctly when Elasticsearch is unavailable on Render.
- [RECOMMENDED] Create a Cloudflare R2 bucket and integrate it for voice audio permanent storage with signed URL access.
- [RECOMMENDED] Do a full end-to-end test on Render Starter with all env vars before presenting to client.